

Computer Networks Lecture 03: Networking Devices and Network Topologies

1 Introduction

A computer network requires different devices to allow computers and other systems to communicate.

For example, computers in a local network may need to connect to each other through a switch. A router may be required when the local network needs to communicate with another network or with the Internet.

Different networking devices perform different functions. Some devices mainly connect computers, while others forward data between networks.

Apart from networking devices, the physical and logical arrangement of devices in a network is also important.

This arrangement is called network topology.

In this lecture, we will study common networking devices and major network topologies.

2 Networking Devices

Networking devices are hardware devices used to connect computers, networks, and other communication systems.

Common networking devices include:

- Hub
- Switch
- Router
- Bridge
- Repeater
- Gateway
- Wireless Access Point
- Modem

Each device performs a different function.

3 Hub

3.1 Definition

A hub is a basic networking device used to connect multiple devices in a local network.

A hub operates mainly at the Physical Layer of the OSI model.

It receives a signal on one port and repeats that signal to the other connected ports.

3.2 Actual Working

1. Receives an incoming signal.
2. Repeats the signal to multiple ports.
3. Allows connected devices to receive the transmitted signal.

A hub does not normally examine the destination MAC address.

Because it sends the signal to multiple ports, unnecessary traffic can be created.

Older Ethernet networks used hubs, but modern networks generally use switches instead.

3.3 Example

Suppose four computers are connected to a hub.

When Computer A sends data, the hub repeats the signal to the other connected ports. The other computers receive the signal and determine whether the data is intended for them.

4 Switch

4.1 Definition

A switch is a networking device commonly used to connect devices within a Local Area Network.

A typical Ethernet switch operates mainly at the Data Link Layer.

It uses MAC addresses to make forwarding decisions.

4.2 Actual Working

1. Learns the MAC addresses associated with its ports.
2. Examines the destination MAC address of an Ethernet frame.
3. Forwards the frame through the appropriate port when possible.

A switch can significantly reduce unnecessary traffic compared with a traditional hub.

A switch can also support full-duplex communication, allowing devices to transmit and receive simultaneously.

Modern switches may also provide additional features such as VLANs, link aggregation, security controls, and management functions.

4.3 Example

Suppose five computers are connected to a switch.

Computer A wants to communicate with Computer D.

The switch examines the destination MAC address and forwards the frame toward the port associated with Computer D instead of simply repeating the frame to every port.

5 Router

5.1 Definition

A router is a networking device used to connect different networks and forward packets between them.

Routers operate primarily at the Network Layer of the OSI model.

They use logical addressing, such as IP addresses, and routing information to determine where packets should be forwarded.

5.2 Actual Working

1. Receives an IP packet.
2. Examines the destination IP address.
3. Forwards the packet toward an appropriate next hop or destination network.

A router can connect a local network to another network, such as the Internet.

Routers maintain routing information that helps them determine where packets should go.

Many home routers also provide additional functions such as NAT, DHCP, firewall features, and wireless access.

5.3 Example

Suppose a home contains several computers and phones connected to a local network.

When a user opens a website on the Internet, the router receives packets from the local device and forwards them toward the Internet.

When the web server sends a response, the router receives the returning packets and forwards them toward the appropriate device in the home network.

6 Bridge

6.1 Definition

A bridge is a networking device used to connect and separate segments of a local network. It operates primarily at the Data Link Layer and uses MAC addresses when making forwarding decisions.

6.2 Actual Working

1. Connects network segments.
2. Examines MAC addresses in frames.
3. Forwards frames between segments when required.

Bridges were commonly used to divide large LANs into smaller segments.

Modern switches perform the functions of bridges and provide many additional ports and features.

6.3 Example

Two Ethernet network segments can be connected using a bridge so that devices on the two segments can communicate.

7 Repeater

7.1 Definition

A repeater is a device that receives a signal and regenerates or retransmits it to extend the physical transmission distance.

It operates at the Physical Layer.

7.2 Actual Working

1. Receives a signal.
2. Regenerates or restores the signal.
3. Transmits the signal toward the next section of the network.

A repeater does not normally make routing or addressing decisions.

7.3 Example

If a signal cannot travel the required distance through a particular cable segment, a repeater can be used to extend the physical network distance.

8 Gateway

8.1 Definition

A gateway is a device or system that connects networks or systems that may use different protocols or communication methods.

The term gateway is used in several contexts, so its exact function depends on the system.

A gateway can translate between different communication systems.

8.2 Actual Working

1. Connects different communication environments.
2. Can translate between different protocols or formats.
3. Allows systems using different technologies to communicate.

8.3 Example

An enterprise system may use a gateway to communicate with another system that uses a different communication protocol.

9 Wireless Access Point

9.1 Definition

A Wireless Access Point, commonly called an access point or AP, provides wireless network connectivity to devices.

It allows wireless devices such as laptops and smartphones to communicate with a wired network or with other wireless devices.

9.2 Actual Working

1. Provides wireless connectivity using radio communication.
2. Connects wireless devices to a local network.
3. Transmits and receives wireless network frames.

9.3 Example

In a university, an access point can allow students to connect their phones and laptops to the university network using Wi-Fi.

10 Modem

10.1 Definition

A modem is a device that provides communication between a local network and a service provider's communication system.

The exact technology used by a modem depends on the Internet connection.

Modern devices often combine modem, router, switch, and wireless access point functions into one device.

10.2 Example

A home Internet device provided by an ISP may contain a modem for the Internet connection, a router for network routing, a switch for wired connections, and Wi-Fi functionality.

11 Network Topology

11.1 Definition

Network topology describes how devices and communication links are arranged in a network.

Topology can describe the physical arrangement of cables and devices or the logical arrangement of communication.

Common network topologies include:

- Bus topology
- Star topology
- Ring topology
- Mesh topology
- Tree topology
- Hybrid topology

12 Bus Topology

12.1 Definition

In a bus topology, multiple devices share a common communication cable or backbone. All devices are connected to the same main communication path.

12.2 Working

When a device transmits data, the signal travels along the shared medium.

The other devices can receive the signal and determine whether the data is intended for them.

Traditional bus Ethernet networks required appropriate termination at the ends of the shared cable.

12.3 Advantages

- Simple design.
- Requires less cable than some other topologies.
- Relatively low initial cost for small networks.

12.4 Disadvantages

- Failure of the main cable can affect the entire network.
- Troubleshooting can be difficult.
- Shared communication can reduce performance.
- Security can be weaker because communication uses a shared medium.

12.5 Example

Older Ethernet networks using coaxial cable are an example of bus topology.

13 Star Topology

13.1 Definition

In a star topology, every device is connected to a central networking device.

The central device is commonly a switch or access point.

13.2 Working

When a computer sends data, the data first reaches the central device.

The central device then forwards the data toward the appropriate destination.

13.3 Advantages

- Easy to install and manage.
- Failure of one individual cable normally affects only one device.
- Easy to expand by connecting additional devices to the central device.
- Modern switched Ethernet networks provide good performance.

13.4 Disadvantages

- Failure of the central device can affect many or all connected devices.
- Requires more cable than a simple bus arrangement.
- The cost of the central device must be considered.

13.5 Example

A typical computer laboratory where every computer is connected to a central switch uses a star topology.

14 Ring Topology

14.1 Definition

In a ring topology, each device is connected to two neighboring devices, forming a circular arrangement.

The communication path forms a ring.

14.2 Working

Data travels around the ring according to the communication method used by the network.

Some ring networks use controlled access methods to determine which device can transmit.

14.3 Advantages

- Provides an organized communication path.
- Can provide predictable access in some implementations.
- Does not require one central device for basic connectivity.

14.4 Disadvantages

- A link or device failure can affect communication in a basic ring.
- Adding or removing devices can be more difficult.
- Troubleshooting can be more complicated.

14.5 Example

Some older network technologies used ring-based arrangements. Ring structures are also used in some specialized and telecommunications networks.

15 Mesh Topology

15.1 Definition

In a mesh topology, devices have multiple connections with other devices.

A full mesh connects every device directly to every other device.

A partial mesh provides multiple connections but does not connect every device to every other device.

15.2 Working

If one communication path becomes unavailable, data may be able to use another available path.

This makes mesh topology useful where redundancy and availability are important.

15.3 Advantages

- High reliability.
- Multiple paths can provide redundancy.
- Failure of one link may not disconnect the entire network.
- Useful for networks where availability is important.

15.4 Disadvantages

- Expensive because many links may be required.
- Installation and maintenance can be complicated.
- A full mesh requires a large number of connections as the number of devices increases.

15.5 Example

Large-scale network backbones can use partial mesh designs so that traffic has alternative paths if one link fails.

16 Tree Topology

16.1 Definition

A tree topology combines characteristics of hierarchical and star arrangements.

Devices are organized in levels, with higher-level devices connecting lower-level devices.

16.2 Working

A central or higher-level device connects to lower-level networking devices, and those devices connect to additional devices.

This creates a hierarchical structure.

16.3 Advantages

- Easy to organize large networks hierarchically.
- Network expansion can be structured.
- Faults can sometimes be isolated to a particular branch.

16.4 Disadvantages

- Failure of an important higher-level device can affect an entire branch.
- Requires more planning than a simple star.
- Higher-level links can become important points of congestion.

16.5 Example

A university network may have a core network connected to departmental switches, with computers connected below those switches.

17 Hybrid Topology

17.1 Definition

A hybrid topology is a network that combines two or more different topology types.

For example, one part of a network may use a star arrangement while another part uses a mesh arrangement.

17.2 Advantages

- Flexible design.
- Can combine the advantages of different topologies.
- Suitable for large and complex networks.

17.3 Disadvantages

- More complex to design.
- Can be more expensive.
- Requires careful network management.

18 Comparison of Network Topologies

Different topologies have different characteristics.

Important factors when selecting a topology include reliability, cost, number of ports or connections, security, scalability, performance, and ease of maintenance.

18.1 Reliability

Reliability describes how well a network continues to operate when failures occur.

A basic bus topology has a major dependency on the shared backbone. If the main cable fails, communication can be seriously affected.

A star topology is more reliable for individual cable failures because failure of one connection normally affects only one device. However, failure of the central switch can affect many devices.

A mesh topology provides high redundancy because multiple paths can exist between devices.

Therefore, mesh designs are commonly useful when high availability is required.

18.2 Cost

Cost includes equipment, cables, installation, maintenance, and expansion.

Bus topology can have a relatively low cable requirement, although traditional bus networks are uncommon today.

Star topology requires a central switch and separate connections to devices.

Mesh topology can be expensive because it requires many communication links.

As the number of devices increases, the cost difference between these topologies can become significant.

18.3 Number of Ports and Connections

The number of required ports or physical connections depends on the topology and the networking devices being used.

In a star topology, each device generally needs one connection to the central switch.

The switch therefore needs enough ports for the connected devices.

In a full mesh topology, every device needs connections to every other device.

If there are only a few devices, this may be manageable. However, as the number of devices increases, the number of connections increases rapidly.

18.4 Security

Topology itself does not completely determine network security.

Security also depends on switches, routers, encryption, authentication, access control, firewalls, network configuration, and other security mechanisms.

However, topology can influence the security design.

A shared bus makes the communication medium common to multiple devices.

A switched star network can provide better control over individual connections, especially when combined with VLANs, authentication, and access control.

Mesh networks can provide multiple paths, but multiple paths also require proper security management.

18.5 Scalability

Scalability means how easily a network can grow.

Star topology is generally easy to expand by connecting additional devices to available switch ports.

Bus topology is less suitable for modern large networks.

Mesh topology provides redundancy but becomes increasingly complex as more devices and links are added.

Hierarchical tree designs can be useful for organizing large networks.

18.6 Maintenance

Maintenance refers to how easily a network can be monitored, repaired, and managed.

Star networks are generally easier to troubleshoot because each device has a separate connection to the central device.

Bus networks can be more difficult to troubleshoot because multiple devices share the same main communication path.

Mesh networks can provide redundancy but may require more complex monitoring and configuration.

19 General Comparison

A simplified comparison can be made as follows.

Bus topology has relatively low physical complexity but has a major dependency on the shared backbone.

Star topology provides a practical balance of cost, performance, management, and expansion and is widely used in modern LANs.

Ring topology provides an organized communication path but basic ring designs can be affected by link or device failures.

Mesh topology provides high redundancy and reliability but can require more connections and higher cost.

Tree topology is useful for hierarchical networks and large organizations.

Hybrid topology provides flexibility by combining multiple topology types.

20 Conclusion

Networking devices and network topologies are two important parts of computer networking.

Devices such as hubs, switches, routers, bridges, repeaters, gateways, access points, and modems perform different networking functions.

A hub mainly repeats signals to connected ports. A switch forwards frames using local addressing information. A router forwards packets between networks using logical addressing and routing information.

Network topology describes how devices and communication links are arranged.

Common topologies include bus, star, ring, mesh, tree, and hybrid.

When selecting a topology, factors such as reliability, cost, number of ports or connections, security, scalability, performance, and maintenance requirements should be considered.

21 References

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